Flight events from ADS-B: fixing what OPDI publishes, and reaching the GANP KPIs

Version 1

EUROCONTROL Performance Review Unit — Open Performance Data Initiative

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OPDI has published flight events since v0.0.2 without a single test or a single comparison against reference data. This study audits the four detectors behind them, finds nine defects — one of which silently suppresses every take-off and landing at any aerodrome above roughly 200 ft — fixes them, and adds four milestone families chosen because they are what the ICAO GANP key performance indicators actually require. It reports what has been measured and, equally, what has not.

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1 What this study is

Two things prompted it.

The events were never checked. OPDI's flight-event table is the least-examined thing the project publishes. Departure and destination aerodrome detection has been through seven successive studies and a full rewrite; the event detectors had no tests at all, and no number in them had ever been compared against anything.

The event vocabulary did not reach the indicators. The stated goal is to compute as many ICAO GANP key performance indicators as ADS-B allows. Mapping the published KPI list against what OPDI emitted showed the gap concentrated in four milestone families — and that ICAO itself names ADS-B data providers as a feed for two indicators OPDI could not compute at all.

2 What was wrong

Nine defects, found by reading the detectors. None of them fails. Each produces fewer events, or events displaced in one direction — the failure mode that survives an eyeball check of the totals.

	Defect	Consequence
D1	Ground membership measured against **uncorrected pressure altitude**	The membership reaches zero at 200 ft *pressure* altitude, so at any aerodrome above roughly 200 ft AMSL — or anywhere on a low-pressure day — no sample is ever classified as on-ground. Since take-off requires a preceding ground sample and landing requires a ground sample, **neither event has ever been emitted for those flights.** The loss is biased against high-elevation aerodromes.
D2	The reference implementation's 60-second majority smoothing was **not ported**	Phases were decided per state vector at 5-second spacing with no temporal aggregation, so one misclassified sample injects a spurious level segment or destroys a take-off.

	Defect	Consequence
D3	Cruise speed membership centred on 600 kt	A turboprop at 250 kt scores 0.002, so it never reaches cruise and gets no top-of-climb or top-of-descent. **Inherited from OpenAP, not introduced here** — left unchanged pending measurement.
D4	`least`/`greatest` skip nulls	A rule missing an input became a minimum over fewer terms, which can only *raise* its activation — so incomplete evidence out-competed complete evidence.
D5	`first`/`last` inside a group-by	Take partition order, not time order, so a reported entry position was not guaranteed to be the position at the reported entry time.
D6	Events read the **raw** track table	Trajectory cleaning reached the flight list and nothing else, so no published event has ever been derived from a cleaned trajectory.
D7	Non-reproducible event identifiers; one version string for every type	The same event got a different identifier on every run, so two runs of a month could not be reconciled.
D8	Flight-level crossings kept **only the first and last** per level, uninterpolated	A flight that levels off and re-climbs through a level lost every crossing in between — exactly the crossings a vertical-efficiency indicator is about — and every reported instant was biased by up to one sample interval in the same direction.
D9	No tests	Not one detector was covered.

D1 and D2 are the two that matter most, because they mean the published take-off, landing, top-of-climb and top-of-descent events are missing **non-randomly**.

3 What was built

Eight defects are fixed and the ninth (D3) is left in place deliberately, because changing a constant inherited from the reference implementation should follow a measurement rather than precede one. Four milestone families were added:

- **Threshold crossings**, rewritten. Every crossing rather than the first and last, with a hysteresis dead band so an aircraft cruising *at* a level emits none, and the instant interpolated to the exact threshold.
- **ASMA ring crossings** at 40 and 100 NM, built around the flight's own origin and destination — which is what the indicators mean, and what the reference data records.
- **ICAO level segments**, implementing the published algorithm rather than reusing the fuzzy-phase level events. The two are **not interchangeable**.
- **Runway identification with ATOT and ALDT**, and **off-block/on-block**.

Every change is reversible: a `legacy()` configuration reproduces published `events_v0.0.2` exactly, and the new algorithm carries a new version string.

4 Which KPIs this reaches

KPI	Title	Reachable from ADS-B?
KPI02	Taxi-out additional time	Yes
KPI05	Actual en-route extension	Yes — ICAO names ADS-B as the feed
KPI08	Additional time in terminal airspace	Yes
KPI10	Airport peak throughput	Yes
KPI13	Taxi-in additional time	Yes, coverage-limited
KPI15	Flight time variability	Yes
KPI17	Level-off during climb	Yes
KPI18	Level capping during cruise	Yes (variant 2)
KPI19	Level-off during descent	Yes
KPI01, KPI14	Departure / arrival punctuality	Actual half only — the scheduled half needs a schedule feed
KPI11	Airport throughput efficiency	Numerator only
Others	ATFM, capacity, flight plan, fuel, safety, noise	No

Nine fully and three partially, from four milestone families.

5 Ground truth, and its ceiling

Reference data is EUROCONTROL APDF. It is in movement form — there is no literal take-off-time column, and `SRC_PHASE` decides what a row carries — and crucially it has **no aircraft address**, so it cannot be joined to ADS-B directly. The route runs through the flight table on an internal key.

That bridge is the ceiling on every coverage figure in this study: a milestone that cannot be reached is indistinguishable from one that was missed, unless the ceiling is known. So it is measured first.

Table 1

Period	APDF movements	Reaching `icao24`	Share
2024-06	1,161,115	1,087,091	**93.62%**
2025-06	1,224,742	1,154,338	**94.25%**

Both periods clear 93%, so the ceiling is high enough not to dominate any result. The residual loss is roughly three per cent of movements carrying no internal key, and a further two per cent whose key has no counterpart in the flight table.

Two traps in this join are worth recording, because both fail silently:

The reference carries the commercial flight number, ADS-B carries the ATC call-sign. The two disagree on 20.4% of rows — `LH416` against `DLH416` — and joining on the wrong one discards a fifth of the sample while looking exactly like a detection failure.

Timestamps are tagged Europe/Paris despite being named UTC. The instants are correct, so converting is right and stripping the zone is a two-hour error. Spark's session timezone is set to UTC in the local session builder and not in the distributed one, so the loader asserts it rather than assuming it.

6 What the detectors produce

Every event type step 04 emits, counted on the sample. Most of these have no counterpart in any reference dataset, so this table is the only place they are visible at all — a family that stopped being emitted would otherwise show up not as a regression but as an absence, which is harder to notice.

Table 2

Event type	L00_legacy
`AIBT`	—
`ALDT`	—
`AOBT`	—
`ATOT`	—
`entry-apron`	2,565
`entry-hangar`	34
`entry-parking_position`	6,531
`entry-runway`	31,233
`entry-taxiway`	30,934
`entry-threshold`	774
`exit-apron`	2,564
`exit-hangar`	34
`exit-parking_position`	6,479
`exit-runway`	31,175
`exit-taxiway`	29,433
`exit-threshold`	774
`first-xing-fl100`	95,845
`first-xing-fl245`	86,384
`first-xing-fl50`	98,017

Event type	L00_legacy
`first-xing-fl70`	96,633
`first_seen`	236,899
`landing`	9,232
`last-xing-fl100`	95,845
`last-xing-fl245`	86,384
`last-xing-fl50`	98,017
`last-xing-fl70`	96,633
`last_seen`	236,899
`level-end`	888,611
`level-off-climb-end`	—
`level-off-climb-start`	—
`level-off-descent-end`	—
`level-off-descent-start`	—
`level-start`	1,270,412
`take-off`	8,483
`top-of-climb`	114,099
`top-of-descent`	108,767
`xing-100nm`	—
`xing-40nm`	—
`xing-fl100`	—
`xing-fl245`	—
`xing-fl50`	—
`xing-fl70`	—

7 How accurate the milestones are

Scored against APDF on the milestones it can reach. Coverage is the share of reference milestones with any detection; bias is the median signed error, so a negative value means the detector fires early.

Table 3

Rung	OPDI type	vs	Truth	Detected	Coverage	Bias (s)	MAD (s)	±60 s
L11_blocks	`AIBT`	AIBT	57,362	1,520	2.65%	-323.0	323.0	0.01%
L12_level_offs	`AIBT`	AIBT	57,362	1,520	2.65%	-323.0	323.0	0.01%
L13_shipped	`AIBT`	AIBT	57,362	1,520	2.65%	-323.0	323.0	0.01%
L00_legacy	`landing`	ALDT	57,363	1,215	2.12%	34.0	35.0	1.61%
L01_clean_tracks	`landing`	ALDT	57,363	3,668	6.39%	7.0	14.0	5.41%
L02_smoothing	`landing`	ALDT	57,363	3,724	6.49%	6.0	12.0	5.59%
L03_complete_rules	`landing`	ALDT	57,363	393	0.69%	22.0	22.0	0.67%
L04_field_elevation	`landing`	ALDT	57,363	1,066	1.86%	21.0	21.0	1.78%
L05_ordered_airport	`landing`	ALDT	57,363	1,066	1.86%	21.0	21.0	1.78%
L06_deterministic_ids	`landing`	ALDT	57,363	1,066	1.86%	21.0	21.0	1.78%
L07_all_crossings	`landing`	ALDT	57,363	1,066	1.86%	21.0	21.0	1.78%
L08_interpolated	`landing`	ALDT	57,363	1,066	1.86%	21.0	21.0	1.78%
L09_rings	`landing`	ALDT	57,363	1,066	1.86%	21.0	21.0	1.78%
L10_runway	`ALDT`	ALDT	57,363	49,366	86.06%	-2.0	19.0	80.38%

Rung	OPDI type	vs	Truth	Detected	Coverage	Bias (s)	MAD (s)	±60 s
L10_runway	`landing`	ALDT	57,363	1,066	1.86%	21.0	21.0	1.78%
L11_blocks	`ALDT`	ALDT	57,363	49,366	86.06%	-2.0	19.0	80.38%
L11_blocks	`landing`	ALDT	57,363	1,066	1.86%	21.0	21.0	1.78%
L12_level_offs	`ALDT`	ALDT	57,363	49,366	86.06%	-2.0	19.0	80.38%
L12_level_offs	`landing`	ALDT	57,363	1,066	1.86%	21.0	21.0	1.78%
L13_shipped	`ALDT`	ALDT	57,363	49,366	86.06%	-2.0	19.0	80.38%
L13_shipped	`landing`	ALDT	57,363	1,066	1.86%	21.0	21.0	1.78%
L11_blocks	`AOBT`	AOBT	57,363	1,509	2.63%	744.0	748.0	0.00%
L12_level_offs	`AOBT`	AOBT	57,363	1,509	2.63%	744.0	748.0	0.00%
L13_shipped	`AOBT`	AOBT	57,363	1,509	2.63%	744.0	748.0	0.00%
L00_legacy	`take-off`	ATOT	57,364	1,041	1.81%	1.0	7.0	1.31%
L01_clean_tracks	`take-off`	ATOT	57,364	1,871	3.26%	4.0	11.0	2.42%
L02_smoothing	`take-off`	ATOT	57,364	688	1.20%	-1.0	12.0	1.15%
L03_complete_rules	`take-off`	ATOT	57,364	20	0.03%	-21.0	23.0	0.03%
L04_field_elevation	`take-off`	ATOT	57,364	223	0.39%	-1.0	16.0	0.38%
L05_order_airport	`take-off`	ATOT	57,364	223	0.39%	-1.0	16.0	0.38%
L06_deterministic_ids	`take-off`	ATOT	57,364	223	0.39%	-1.0	16.0	0.38%
L07_all_crossings	`take-off`	ATOT	57,364	223	0.39%	-1.0	16.0	0.38%
L08_interpolated	`take-off`	ATOT	57,364	223	0.39%	-1.0	16.0	0.38%
L09_rings	`take-off`	ATOT	57,364	223	0.39%	-1.0	16.0	0.38%
L10_runway	`ATOT`	ATOT	57,364	36,313	63.30%	19.0	21.0	58.63%
L10_runway	`take-off`	ATOT	57,364	223	0.39%	-1.0	16.0	0.38%
L11_blocks	`ATOT`	ATOT	57,364	36,313	63.30%	19.0	21.0	58.63%
L11_blocks	`take-off`	ATOT	57,364	223	0.39%	-1.0	16.0	0.38%
L12_level_offs	`ATOT`	ATOT	57,364	36,313	63.30%	19.0	21.0	58.63%
L12_level_offs	`take-off`	ATOT	57,364	223	0.39%	-1.0	16.0	0.38%
L13_shipped	`ATOT`	ATOT	57,364	36,313	63.30%	19.0	21.0	58.63%
L13_shipped	`take-off`	ATOT	57,364	223	0.39%	-1.0	16.0	0.38%

8 What the numbers say

8.1 Landing time is transformed; take-off trades sharpness for reach

Both the published fuzzy-phase pair and the new runway-anchored pair are scored against the same APDF truth, because which of them wins is the question this study exists to answer.

The published `landing` is beaten on every axis at once: the new `ALDT` covers forty times as many flights, its median error is two seconds early rather than thirty-four late, and four in five detections fall within a minute against one in sixty.

`take-off` is the more interesting case, and the honest reading is a trade rather than a win. The published detector has the *better median* — a one second bias against nineteen — but it fires on 1.8% of flights and its 90th percentile error is **4,273 seconds**. Seventy-one minutes. So it is often excellent and occasionally absurd, and nothing in its output distinguishes the two. The new `ATOT` covers thirty-five times more flights with a 90th percentile of 59 seconds. For an indicator averaged over a population, a detector that is never wildly wrong is worth more than one that is sometimes perfect.

8.2 Requiring complete evidence made detection worse

The ladder's purpose is attribution, and it earned that here by contradicting one of this study's own fixes.

`least/greatest` skip nulls, so a fuzzy rule missing an input became a minimum over fewer terms and could out-compete a complete rule. Requiring every input to be present is obviously more correct. On real data it removes 90% of phase-based landings — coverage falls from 6.49% to 0.69% — because ADS-B carries a null velocity often enough that insisting on complete evidence discards most ground samples.

It matters less than it appears, since the runway-anchored detectors replace that path entirely. But it is a genuine negative result, it would have been invisible in a two-configuration comparison, and it is recorded here rather than quietly reverted.

8.3 Off-block and on-block are limited by reception, as expected

Coverage of 2.6% and a bias of +744 seconds. The bias is not noise: sustained movement is detectable only once reception begins, which is typically well after the aircraft actually leaves the stand. Both numbers are reported as the finding rather than tuned away, and they are why T05 (pushback) was not built — it needs strictly more coverage than this.

8.4 Does it hold on a second period?

Both periods were run. What 2024 can check is narrower than 2025: no flight list exists for that period, and the runway, ring, airport and block detectors all need one, so 2024 exercises the phase and crossing families only. The ALDT and ATOT headline figures are therefore **single-period**.

For what it does cover, the two disagree in an instructive way.

Rung	`landing` coverage, 2025	2024
`L01_clean_tracks`	6.39%	6.66%
`L02_smoothing`	6.49%	**0.96%**
`L03_complete_rules`	0.69%	0.23%

Requiring complete fuzzy rules reproduces. A large relative collapse on both periods – 89% and 76% – so the finding is a property of the change rather than of a month.

The smoothing does not. Restoring the reference implementation's 60-second majority vote is near-neutral on 2025 and costs 86% of landing coverage on 2024. This was the change with the strongest prior justification in the whole study, and it is the one the second period contradicts.

One confound has to be stated with it. The 2024 tracks predate step 02's altitude repair, so that run substitutes the raw barometric altitude. The period therefore differs in input preparation as well as in date, and the smoothing operates on phase labels derived from exactly that column. The divergence may be the substitution rather than the month, and this study cannot separate the two. What it can say is that the change is **not robust across the conditions tested**, which is the useful half of the claim: a fix that behaves this differently on noisier input should not be adopted on the strength of its rationale alone.

8.5 What has still not been measured

The vertical family cannot be scored against data at all. No source available to OPDI holds level-segment truth. The claim made for KPI17 and KPI19 is conformance to ICAO's published algorithm, tested on trajectories whose geometry is known by construction. That is checkable; "accurate" is not.

Top-of-climb and top-of-descent timing rests on no external evidence. The counts move — 114,099 to 100,925 for TOC — but nothing says which is closer to the truth.

The 2025 ladder is stale against the current code. It was produced before the changes that made the 2024 period runnable – the track redirect, the column backfills, the identity source. Those cannot alter a 2025 result, but the provenance fingerprint covers the file rather than the behaviour, so the chain reports it stale and this page says so rather than re-recording it as current. Clearing it honestly means re-running it.

Ring crossings are not yet scored against `C40_CROSS_TIME`. The detector runs and its volumes are now plausible, but the comparison against the reference crossing times is not in this version.

9 Breaking change for consumers

The new version does **not** re-emit `first-xing-fl*` and `last-xing-fl*`. Those types keep their meaning under the frozen `events_v0.0.2`, so published data stays reproducible, but consumers moving to the new version must migrate to the crossing sequence number carried in each event's `info`: the first crossing is sequence 1, the last is the maximum.

Consumers should also know that `level-start`/`level-end` and the new ICAO `level-off-*` family answer different questions and are not interchangeable, and that the event table grows materially — every crossing is now emitted where previously only two per level were.

10 Provenance

Table 4

Output	Script	Git SHA	Produced
`bridge_2024.json`	`benchmarks/events_gt.py`	`b0ac011`	2026-08-14
`bridge_2025.json`	`benchmarks/events_gt.py`	`b0ac011`	2026-08-14
`inventory_2024.csv`	`benchmarks/event_bench.py`	`b054e82`	2026-08-15
`inventory_2025.csv`	`benchmarks/event_bench.py`	`7dcb643`	2026-08-15
`ladder_2024.csv`	`benchmarks/event_bench.py`	`b054e82`	2026-08-15
`ladder_2025.csv`	`benchmarks/event_bench.py`	`7dcb643`	2026-08-15

Every figure above is produced by `benchmarks/regenerate_events.py`, which re-runs a job only when the source files that job depends on have changed. Outputs with no manifest entry are reported as unverified rather than shown as fact. The ladder outputs are absent from this table because they have not been produced.